Names: Jodhvir Singh, Vraj Desai, Rayyan Dulvi, Chase Sims

ENGR 102 Lab 5A DOC

Variables Needed:
(Also, what names should I use for the variables?)

gender, age, isSmoker, systolic_bp, hdl, is_bp_Treated, total_cholesterol, points, risk_percent

Steps:

1. Ask the user for all the information needed.
Gender
- Age
- Smoking status
- Systolic blood pressure
- HDL cholesterol
- If blood pressure is treated
- Total cholesterol

2. Use the person's gender to choose the correct chart.

3. Use the person's age to find their age points.

4. Use their age and total cholesterol to find their cholesterol points.

5. Use their HDL level to find their HDL points.

6. Use their blood pressure and treatment status to find their blood pressure points.

7. Use their smoking status and age to find their smoking points.

8. Add all the points together.

9. Use the total points to find their 10-year risk percentage.

10. Show the total points and risk percentage to the user.

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Test Cases:

Typical Cases:

What is Being Tested	Input	Expected Output
Age Points	Male, Age 45	3 Age Points
Age Points	Female, Age 65	12 Age Points
Cholesterol Points	Male, Age 45, total cholesterol 220 mg/dl	5 Cholesterol Points
Cholesterol Points	Female, Age 55, total cholesterol 250 mg/dl	5 Cholesterol Points
Smoking Points	Male, age 35, smoker	8 Smoking Points
Smoking Points	Male, age 60, nonsmoker	0 Smoking Points
Smoking Points	Female, age 45, smoker	7 Smoking Points
HDL Points	Female, HDL 55 mg/dl	0 HDL Points
BP Points	Male, untreated systolic BP 135 mmHg	1 BP Point
BP Points	Female, untreated systolic BP 125 mmHg	1 BP Point
BP Points	Male, treated systolic BP 145 mmHg	2 BP Points
Entire Program	Male, age 45, cholesterol 200, nonsmoker, HDL 55, untreated systolic BP 130	9 Total Points, 5 % risk
Entire Program	Female, age 45, cholesterol 200, nonsmoker, HDL 55, untreated systolic BP 130	11 Total Points, 1 % risk
Entire Program	Female, age 60, cholesterol 200, smoker, HDL 45, untreated systolic BP 140	18 Total Points, 6 % risk
Entire Program	Male, age 50, cholesterol 240, smoker, HDL 40, treated systolic BP 120	15 Total Points, 20 % risk

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Edge Cases:

1. Gender: Male, Age: 20
Output: -9 Age Points

2. Gender: Male, Age: 34
Output: -9 Age Points

3. Gender: Female, Age: 35
Output: -3 Age Points

4. Gender: Female, Age: 79
Output: 16 Age Points

5. Gender: Male, Age: 39, Total Cholesterol: 160 mg/dL
Output: 4 Cholesterol Points

6. Gender: Female, Age: 49, Total Cholesterol: 280 mg/dL
Output: 10 Cholesterol Points

7. Gender: Male, HDL: 60 mg/dL
Output: -1 HDL Point

8. Gender: Female, HDL: 40 mg/dL
Output: 1 HDL Point

9. Gender: Male, HDL: 39 mg/dL
Output: 2 HDL Points

10. Gender: Male, Systolic BP: 120 mmHg, BP Treated: No
Output: 0 BP Points

11. Gender: Female, Systolic BP: 160 mmHg, BP Treated: No
Output: 4 BP Points

12. Gender: Male, Systolic BP: 160 mmHg, BP Treated: Yes
Output: 3 BP Points

13. Gender: Male, Age: 39, Smoker: Yes
Output: 8 Smoking Points

14. Gender: Female, Age: 70, Smoker: Yes
Output: 1 Smoking Point

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15. Gender: Male, Age: 20, Cholesterol: 159, Nonsmoker, HDL: 60, Untreated BP: 119
Output: -10 Total Points, <1% Risk